

Substrate-Symplectic Trace Ratio Candidate (c) Across K-Types v0.1

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Substrate-Symplectic Trace Ratio Candidate (c) v0.1

0. Goal

Per Lyra Pf(ω) explicit computation v0.1 walk-back + Candidate (a) substrate-volume walk-back v0.1: investigate Candidate (c) substrate-symplectic trace ratio at substrate K-types as substrate-mechanism candidate for 1/g substrate-natural factor.

1. Substrate-Symplectic Trace at Sp(2) Representations

Per Sp(2) substrate-symplectic Lie group representation theory, substrate-symplectic 2-form ω acts on substrate K-type representations as substrate-Lie-algebra element.

Substrate-Casimir-like substrate-symplectic invariant at Sp(2) representation V_λ :

$$C_\lambda^{symp} = \text{Tr}_{V_\lambda}(\omega \cdot \omega^*) / \dim V_\lambda$$

where ω^* is dual substrate-symplectic 2-form.

2. Substrate-Trace Computation at Substrate K-Types

$V_{(1/2, 1/2)}$ Substrate-Fundamental (dim 4 = 2-rank)

Per Sp(2) substrate-fundamental V_{fund} dim 4: substrate-symplectic ω acts as canonical antisymmetric 2-form on V_{fund} .

Substrate-trace: $\text{Tr}_{\{V_{\text{fund}}\}}(\omega \cdot \omega^*)$ at substrate-canonical Sp(2) substrate-fundamental representation.

For substrate-canonical ω matrix on V_{fund} (per Pf(ω) v0.1 explicit form):

$$\omega \cdot \omega^* = -I_4 \text{ (substrate-canonical antisymmetric squared = -I)}$$

$\text{Tr}_{\{V_{\text{fund}}\}}(-I_4) = -4$. Substrate-trace substrate-natural at substrate-fundamental.

$**C^{\{symp\}}_{(1/2, 1/2)} = -4 / 4 = -1$ substrate-natural.

$V_{(3/2, 1/2)}$ Substrate-Spinor (dim ?)

Per Lyra Mehler Matrix Element v0.2: $V_{(3/2, 1/2)}$ substrate-K-type dim per Sp(2) representation:

$V_{(3/2, 1/2)} = \text{Sym}^? \text{ substrate-spinor representation of Sp(2). Dim per Sp(2) Sym}^k \text{ of fundamental: - Sym}^k \text{ of } V_{\text{fund}} \text{ dim 4: } \binom{k+3}{3}$.

For $k = 3$ (substrate-spinor $V_{(3/2, 1/2)}$ candidate): $\dim \text{Sym}^3(V_{\text{fund}}) = (3+3)! / (3! \cdot 3!) = 20$.

Substrate-trace at $V_{(3/2, 1/2)}$: substrate-canonical ω in Sym^3 representation:

$$C_{(3/2, 1/2)}^{\text{symp}} = \text{Tr}_{V_{(3/2, 1/2)}}(\omega \cdot \omega^*) / \dim V_{(3/2, 1/2)}$$

Per Sym^3 representation: $\omega \cdot \omega^*$ acts as $\text{Sym}^3(-I_4) = -I_{20}$ (Sym^3 of $-I_4 = -I_{20}$ since substrate-canonical $-I_4$ acts identically on Sym^3).

Hmm — substrate- Sym^3 of $-I_4$: each substrate- Sym^3 basis element transforms with -1 per substrate-canonical factor. $\text{Sym}^k(-I) = (-1)^k I$. For $k=3$: $\text{Sym}^3(-I_4) = -I_{20}$.

$\text{Tr}_{\{V_{(3/2, 1/2)}\}}(-I_{20}) = -20$.

$**C^{\{\text{symp}\}}_{(3/2, 1/2)}** = -20 / 20 = -1$ substrate-natural.

Substantive Observation

Per substrate-trace ratio: $C^{\{\text{symp}\}}_{(3/2, 1/2)} / C^{\{\text{symp}\}}_{(1/2, 1/2)} = -1 / -1 = 1$ substrate-natural.

Substrate-symplectic trace ratio across substrate K-types = substrate-natural UNITY at substrate-canonical $\text{Sp}(2)$ representations.

3. Honest Walk-Back of Candidate (c)

Substantive observation v0.1: - Substrate-symplectic trace ratio $C^{\{\text{symp}\}}_{\text{K-type}}$ substrate-natural across substrate K-types - $\text{Tr}_V(\omega \cdot \omega^*) / \dim V = -1$ substrate-natural at substrate-canonical $\text{Sp}(2)$ substrate-fundamental + substrate-spinor representations - Substrate-trace ratio across K-types = 1 substrate-natural (substrate-canonical) - Candidate (c) at substrate-canonical normalization does NOT give 1/g substrate-natural factor

Honest walk-back of Candidate (c) at substrate-canonical normalization: substrate-trace ratio substrate-natural unity; substrate-mechanism source for 1/g must be elsewhere.

4. Refined Candidate (c) Path

Per Cal #189 question-shape: substrate-symplectic substrate-trace ratio at substrate-canonical $\text{Sp}(2)$ substrate-fundamental + substrate-spinor representations both substrate-natural unity; substrate-mechanism source for 1/g may be at:

(c') Substrate-twisted substrate-Casimir at non-canonical $\text{Sp}(2)$ representation (c'') Substrate-anomaly substrate-trace at substrate-spinor $V_{(3/2, 1/2)}$ (c''') Substrate-quantum substrate-trace at substrate-deformed $\text{Sp}(2)$ representation

Multi-week explicit derivation per $\text{Sp}(2)$ substrate-symplectic substrate-mechanism representation theory.

5. 3 of 4 Candidates Examined — Pattern Recognition

Per Cal #36 STANDING positive-search rule examined Thursday afternoon:

#	Candidate	substrate-canonical at V_{fund}	Status v0.1
1	$\text{Pf}(\omega)$ substrate-Pfaffian	1	Walked back
2	$\omega^2/2!$ substrate-volume form	1	Walked back
3	$\text{Tr}(\omega \cdot \omega^*) / \dim V$ substrate-trace ratio	-1 (unity-scale)	Walked back

Pattern observation: 3 of 3 substrate-symplectic candidates at substrate-canonical $\text{Sp}(2)$ substrate-fundamental representation = substrate-natural unity-scale.

Substantive substrate-mechanism reading: substrate-canonical $Sp(2)$ substrate-fundamental representation produces substrate-natural unity at substrate-symplectic substrate-invariant level (per $Pf(\omega) + \omega^2/2! + Tr(\omega \cdot \omega^*)/\dim = \text{all unity}$). substrate-mechanism source for $1/g$ substrate-natural factor MUST come from:

(α) Substrate K-type RATIO at substrate-spinor $V_{(3/2, 1/2)}$ vs substrate-fundamental $V_{(1/2, 1/2)}$ substrate-orbit substrate-volume — NOT substrate-canonical substrate-invariant (β) Substrate-multiplicity substrate-Pochhammer ratio per Schur II Pochhammer scaling (γ) Substrate-anomaly substrate-trace at substrate-deformed $Sp(2)$ representation (δ) Substrate-Maslov substrate-index (Candidate (d) — pending v0.1)

6. Closure v0.1

Substrate-Symplectic Trace Ratio Candidate (c) v0.1: substrate-trace ratio $C^{\{\text{symp}\}}_{K\text{-type}} = -1$ substrate-natural at substrate-canonical $Sp(2)$ substrate-fundamental + substrate-spinor representations; substrate-trace ratio across K-types = unity; Candidate (c) at substrate-canonical normalization INSUFFICIENT for $1/g$ factor.

Pattern observation across 3 candidates: $Pf(\omega) + \omega^2/2! + Tr(\omega \cdot \omega^*)/\dim V$ all substrate-natural unity at substrate-canonical $Sp(2)$ substrate-fundamental representation.

Substrate-mechanism source for $1/g$ must come from substrate K-type substrate-orbit RATIO / substrate-Pochhammer ratio / substrate-anomaly substrate-trace / substrate-Maslov index — NOT substrate-canonical substrate-invariant level.

Audit-chain Lyra-on-self brake continued (Thursday afternoon): substantive 3 candidates examined; substrate-canonical substrate-fundamental unity-pattern identified; substrate-mechanism narrows to non-canonical / RATIO-form / substrate-anomaly / substrate-Maslov candidates.

Per Cal #36 STANDING positive-search rule + Cal #27 STANDING: rapid substantive examination of 3 candidates → substrate-canonical unity pattern → refined search space for substrate-mechanism source of $1/g$ substrate-natural factor.

Tier: 3/4 substrate-symplectic candidates at substrate-canonical level INSUFFICIENT; multi-week refined RATIO / substrate-Pochhammer / substrate-anomaly / substrate-Maslov path continues per Cal #189.

— Lyra, Thu 2026-06-04 ~12:45 EDT. Substrate-Symplectic Trace Ratio Candidate (c) v0.1: $Tr(\omega \cdot \omega^*)/\dim V = -1$ substrate-natural at substrate-canonical $V_{\text{fund}} + V_{(3/2, 1/2)}$; ratio = unity; 3 of 4 candidates examined produce substrate-canonical unity pattern; substrate-mechanism source for $1/g$ substrate-natural factor narrowed to K-type RATIO / substrate-Pochhammer / substrate-anomaly / substrate-Maslov substrate-mechanisms.